

Lecture 7

Network Management

CoSc2043 – Network and System Administration

Section Overview

- ◆ TCP/IP Basics

- ◆ Routing

- ◆ TCP/IP Configuration

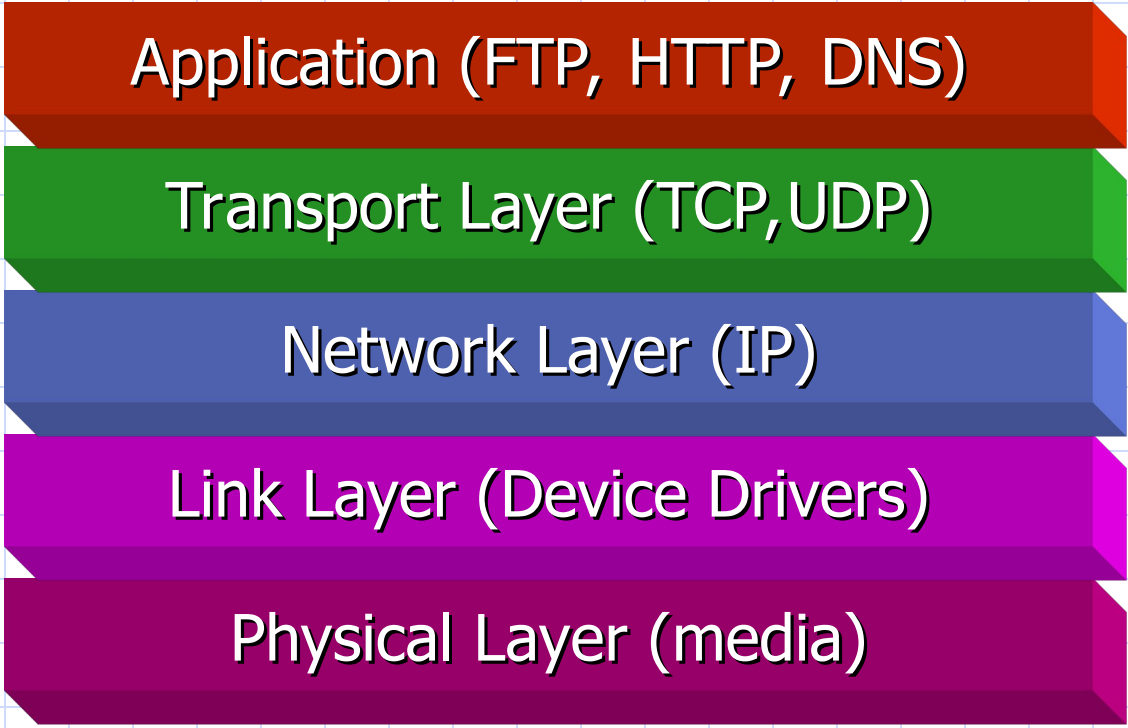
- Configuring IPv4 Network Connectivity
- Configuring IPv6 Network Connectivity
- Implementing Automatic IP Address Allocation
- Troubleshooting Network Issues

- ◆ TCP/IP Network Testing

- ◆ Dynamic Host Config Protocol (DHCP)

- ◆ Wireless Networking

TCP/IP Protocol Stack



A diagram of the TCP/IP Protocol Stack consisting of five stacked, 3D rectangular blocks. From top to bottom, the blocks are: orange (Application), green (Transport Layer), blue (Network Layer), magenta (Link Layer), and dark purple (Physical Layer). Each block has a white border and a slight 3D effect with a shadow on the bottom and right sides.

Application (FTP, HTTP, DNS)

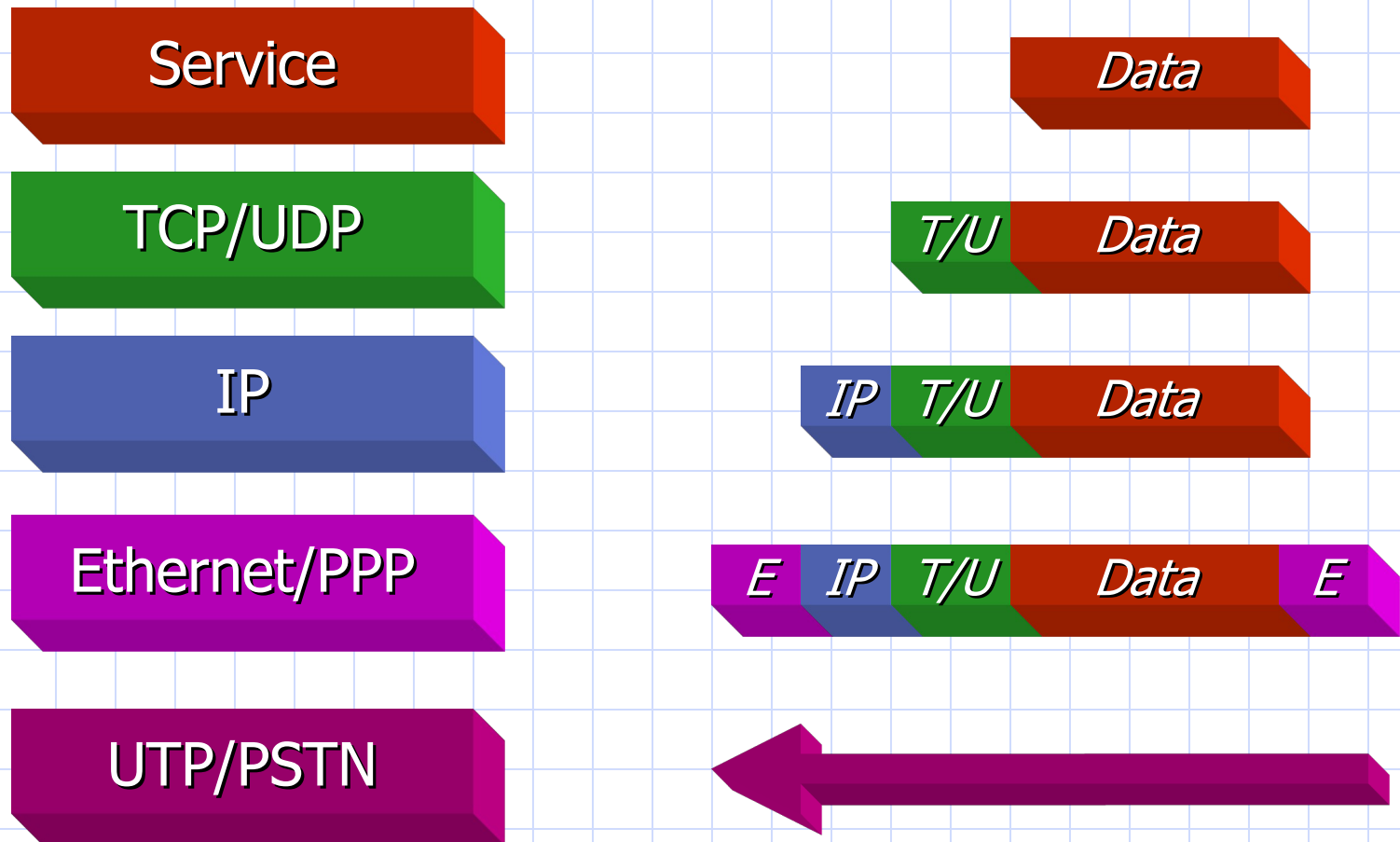
Transport Layer (TCP,UDP)

Network Layer (IP)

Link Layer (Device Drivers)

Physical Layer (media)

TCP/IP Packet Encapsulation



Connecting to a Network

- ◆ Hostname and IP Address assignment
- ◆ Configuration of hardware
- ◆ Default route (gateway) assignment
- ◆ Name Service Configuration
- ◆ Testing and troubleshooting

Configuring IPv4 Network Connectivity

- ◆ What is an IPv4 Address?
- ◆ What is a Subnet Mask?
- ◆ What is a Default Gateway?
- ◆ What are Public and Private IPv4 Addresses?
- ◆ Demonstration: Configuring an IPv4 Address
- ◆ Types of Computer Names
- ◆ Methods for Resolving Computer Names

IP Addresses

- ◆ Unique for each connection (interface)
- ◆ Consists of 4 octets (#.#.#.#)
 - Network portion
 - Host portion
- ◆ Special Addresses
 - Network Address
 - Broadcast Address

IP Address Classes

Class	1 st Byte	Format	Total Hosts
A	0 – 126	N.H.H.H	16 Million
B	128 – 191	N.N.H.H	64 Thousand
C	192 – 239	N.N.N.H	254
D	224 – 239	-	(Multicast)
E	240 – 254	-	(Experimental)

Subnet Masks

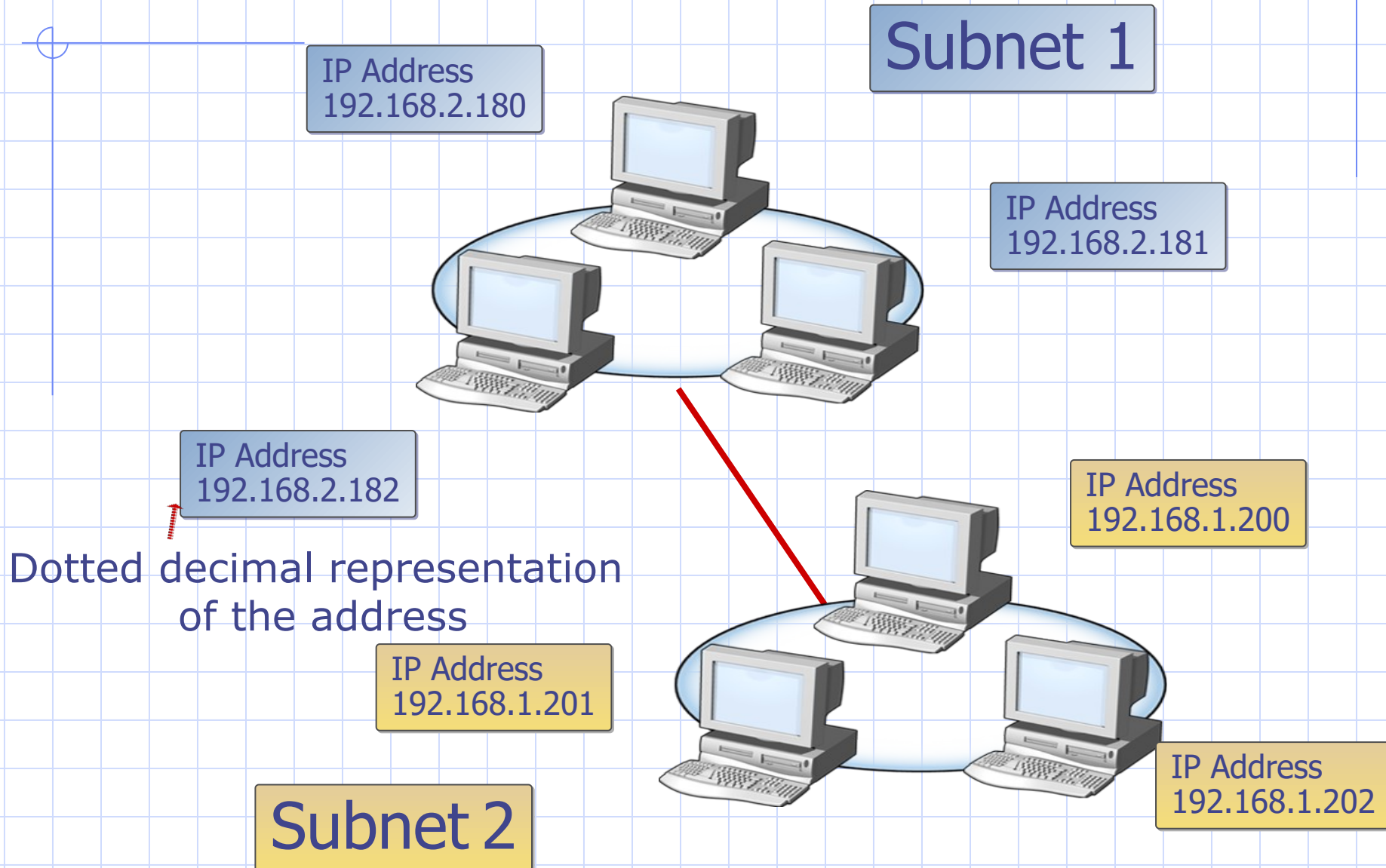
- ◆ Splits networks into subnetworks
- ◆ Separates address into 2 parts
 - 1's – Network Portion
 - 0's – Host Portion
- ◆ Example: Class C Network
 - Address: N.N.N.H
 - Mask: 255.255.255.0 (255 = 11111111)
 - CIDR Notation: N.N.N.H/24

Interface Configuration

- ◆ Hardware to connect to network
- ◆ Common interfaces
 - Ethernet
 - Modem
- ◆ Loopback (lo) Interface
- ◆ **ifconfig** – View/Configure interface
- ◆ **ipconfig** – View interface (Windows)

What is an IPv4 Address?

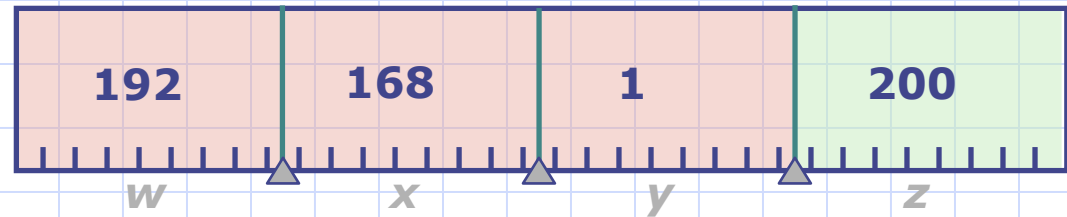
An IPv4 address identifies a computer to other computers on a network.



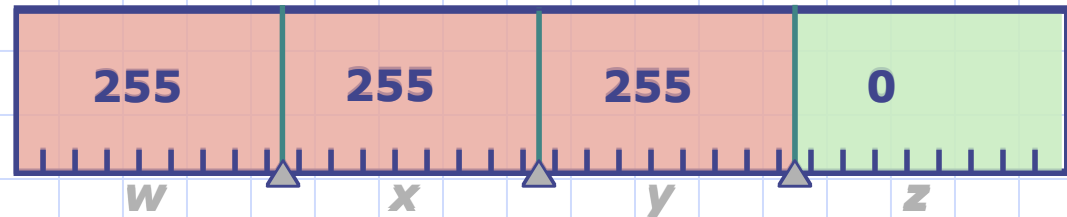
What is a Subnet Mask?

And here is the host ID in the fourth octet. Note that this host ID is 1 of 254 hosts on this specific subnet

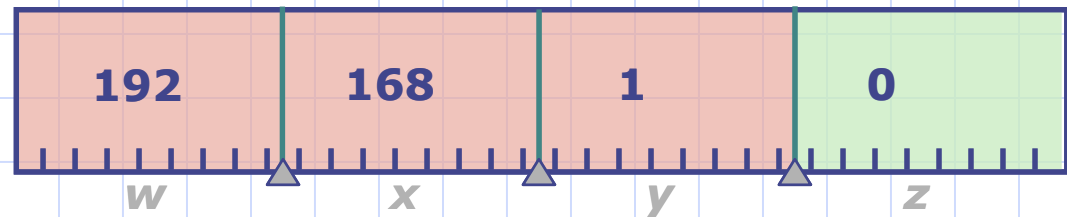
IP address



Subnet mask

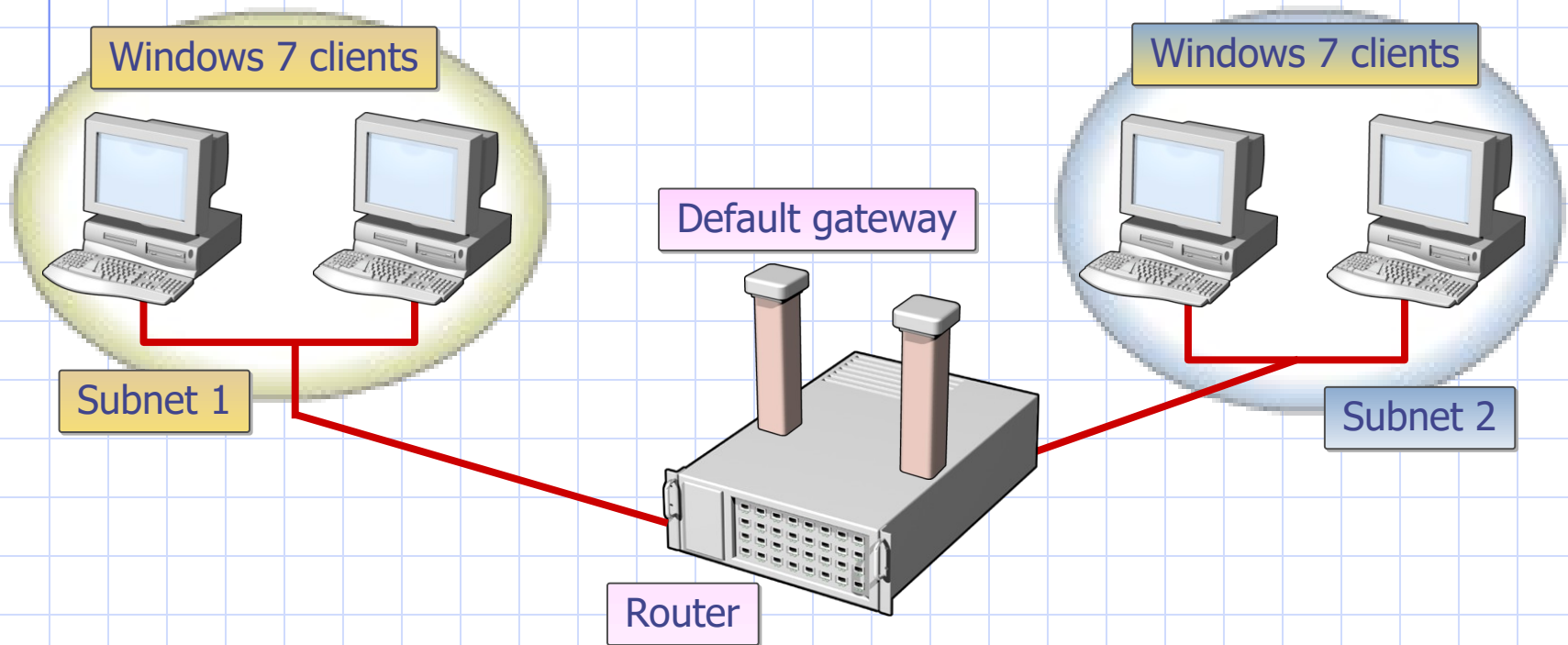


Network ID



What is a Default Gateway?

Use a default gateway when the internal routing table on the host has no information about the destination subnet.



What are Public and Private IPv4 Addresses?

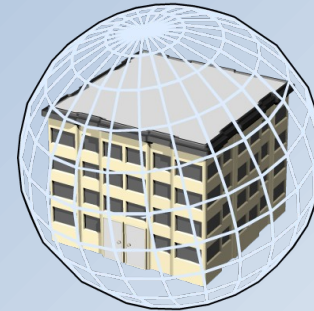
Public

- Required by devices and hosts that connect directly to the Internet
- Must be unique
- Routable on the Internet
- Must be assigned by IANA

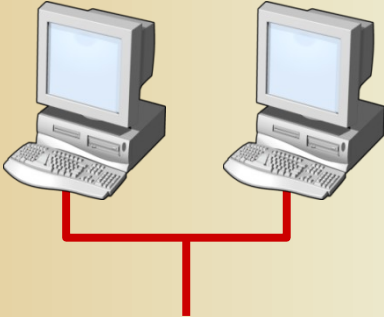


Private

- Non-routable on the Internet
- Can be locally assigned by organization
- Must be translated to access the Internet



Types of Computer Names

Name	Description
 Host name	• Up to 255 characters in length • Can contain alphabetic and numeric characters, periods, and hyphens • Part of FQDN
 NetBIOS name	• Represent a single computer or group of computers • 15 characters used for the name • 16th character identifies service • Flat namespace

Methods for Resolving Computer Names

Domain Name System (DNS)

✓ DNS is a service that manages the resolution of host names to IP addresses

✓ DNS is the name resolution tool of choice for Windows 7

Windows Internet Naming Service (WINS)

✓ WINS is retained for backward compatibility

✓ The Global Domain Names feature can remove the need for WINS

✓ WINS is a NetBIOS name server used to resolve NetBIOS

Configuring IPv6 Network Connectivity

- ◆ Benefits of Using IPv6
- ◆ Windows 7 Support for IPv6
- ◆ What is the IPv6 Address Space?
- ◆ IPv6 Address Types
- ◆ Demonstration: Configuring an IPv6 Address

Benefits of Using IPv6

Benefits of using IPv6 compared to IPv4



Larger address space



More efficient routing



Simpler host configuration



Built-in security



Better prioritized delivery support



Redesigned headers

Support for IPv6

IPv6 is Enabled by Default

Windows 7 uses IPv6 by default to support security needs and additional features

Windows 7 Dual Stack

Windows 7 facilitates the dual stack to use IPv4 and IPv6 simultaneously

Direct Access requires IPv6

Windows 7 clients can use Direct Access which facilitates client computers connecting to the enterprise domain

IPv6 uses Remote Desktop

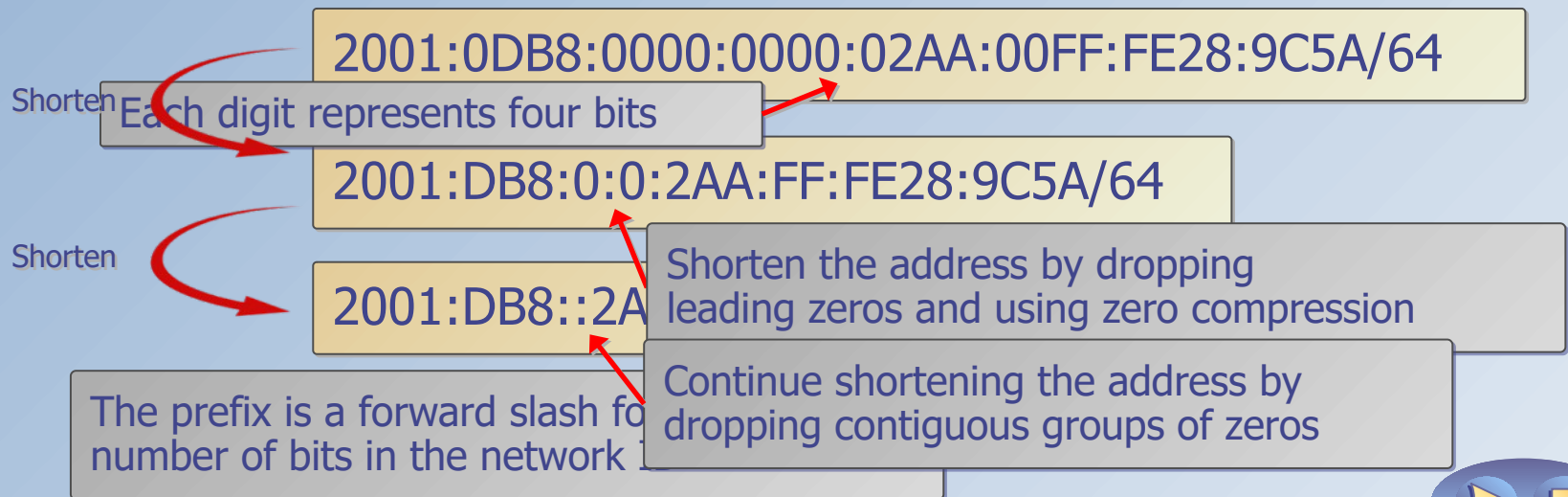
IPv6 supports Windows 7 File Sharing Security and Echo System features such as Remote Access and Direct Access

What is the IPv6 Address Space?

The IPv6 address space:

- Uses 128 bits as compared to 32-bits that the IPv4 address space uses
- Allocates 64-bits for the network ID and 64-bits for the host ID
- Uses a prefix to define the network ID

IPv6 uses hexadecimal notation



IPv6 Address Types

IPv6 Address Types

- Unicast – use for one-to-one communication between hosts
- Multicast – use for one-to-many communication between computers that are defined as using the same multicast address
- Anycast – use for locating services or the nearest router

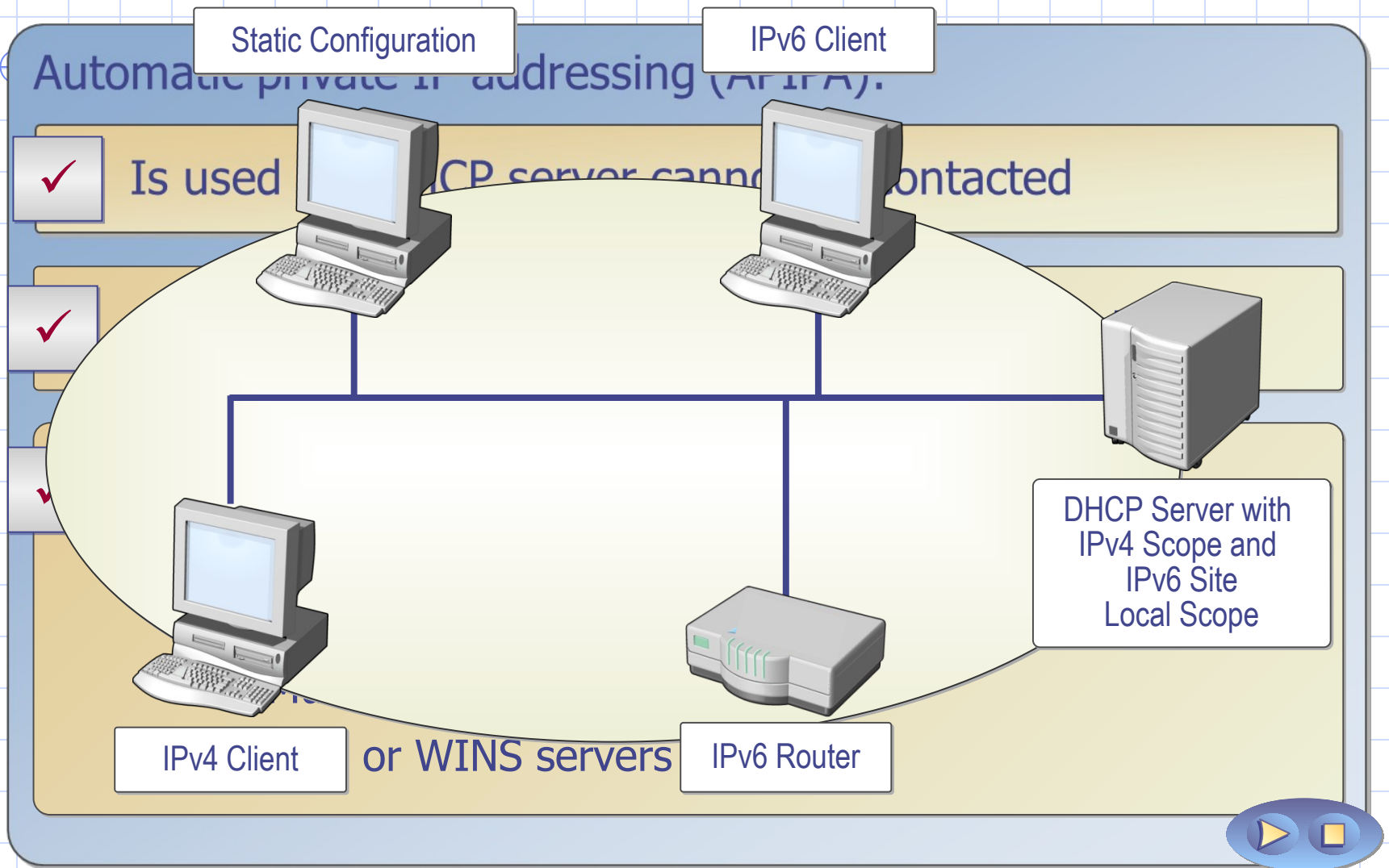
IPv6 Unicast Address Types

- Global Unicast – globally routable and reachable on the IPv6 portion of the Internet
- Link-Local – use when communicating with neighboring hosts on the same link
- Unique Local Unicast – equivalent to IPv4 private address spaces, such as 10.0.0.0/8, and have the prefix FD00::/8

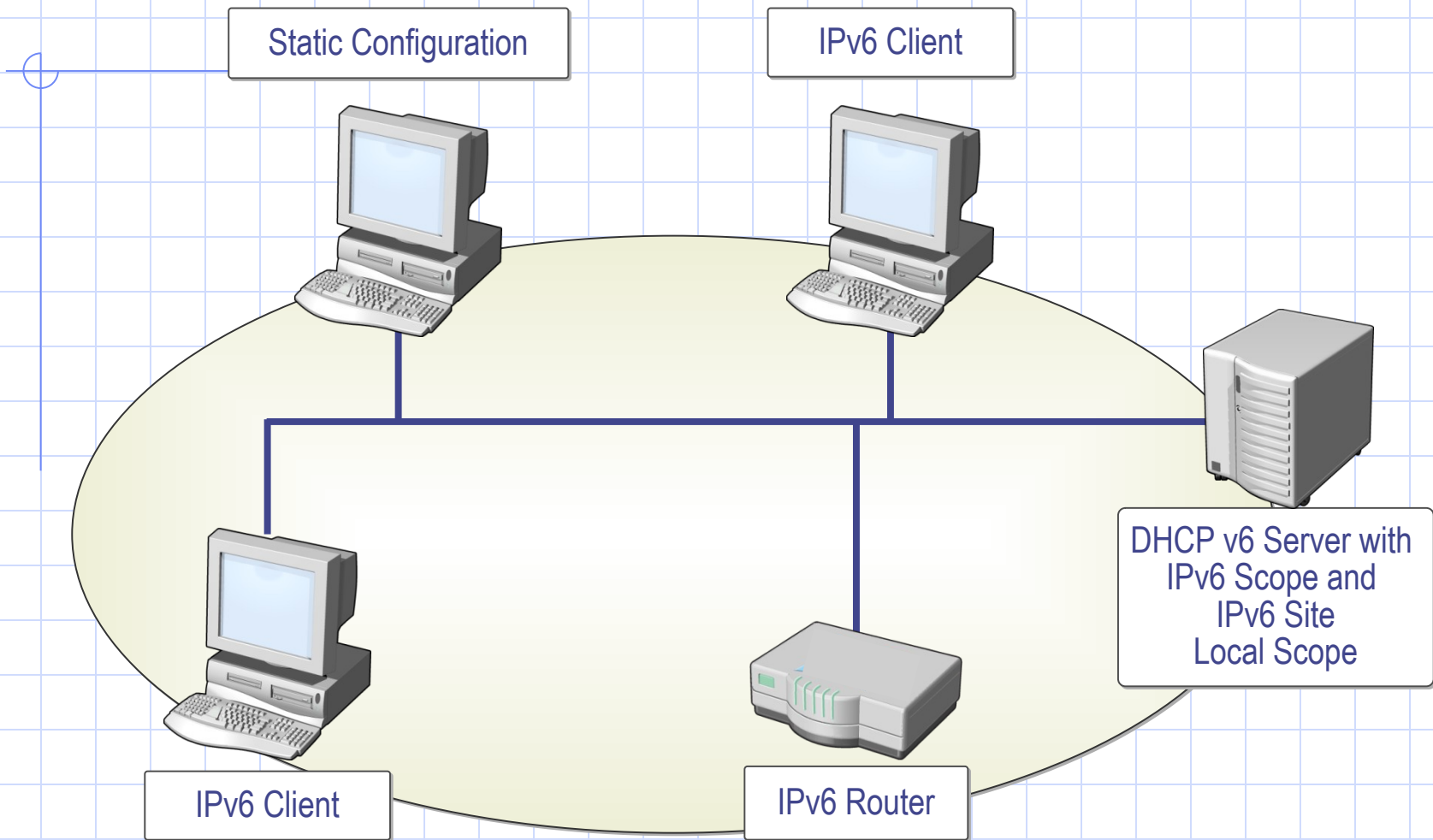
Implementing Automatic IP Address Allocation

- ◆ Automatic IPv4 Configuration Process
- ◆ Automatic IPv6 Configuration Process
- ◆ Demonstration: Configuring a Computer to Obtain an IPv4 Address Dynamically
- ◆ Troubleshooting Client-Side DHCP Issues

Automatic IPv4 Configuration Process



Automatic IPv6 Configuration Process



Troubleshooting Client-Side DHCP Issues

IPConfig is used to display IP configuration information and to release and renew addresses

Option	Description
/all	Displays all IP address configuration information
/release	Releases a dynamic IPv4 address lease
/renew	Renews a dynamic IPv4 address lease

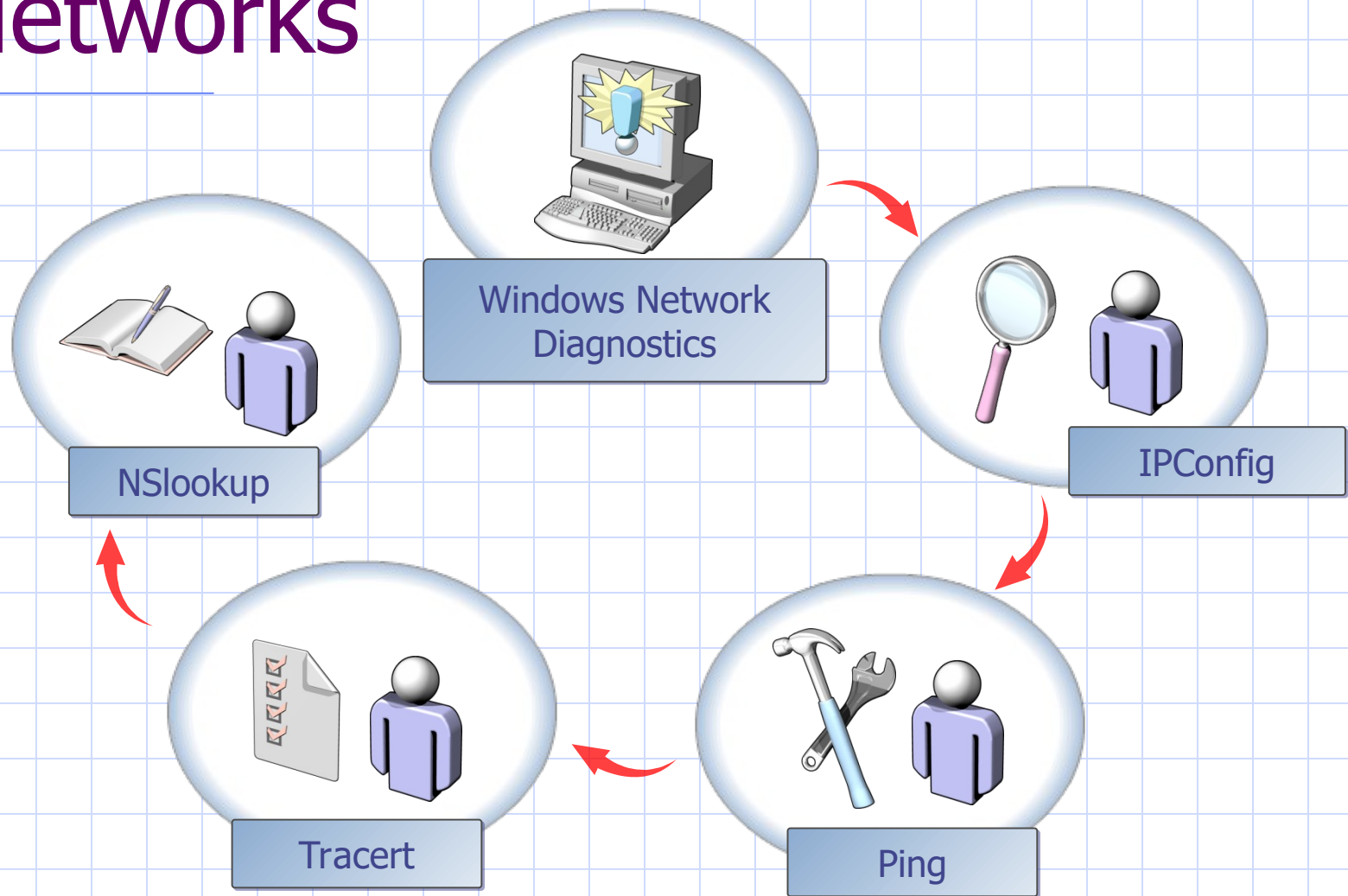
Troubleshooting Network Issues

- ◆ Tools for Troubleshooting Networks
- ◆ Process for Troubleshooting Networks
- ◆ Demonstration: Troubleshooting Common Network-Related Problems

Tools for Troubleshooting Networks

Tool	Purpose
Event Viewer	Enables you to view errors relating to network activity
Windows Network Diagnostics	Helps to diagnose and resolve network problems
IPCONFIG	Displays IP configuration information and controls the DNS resolver cache
PING and PathPING	Verifies basic IP connectivity
TRACERT	Verifies a routing path
NSLOOKUP	Enables testing of name resolution

Process for Troubleshooting Networks

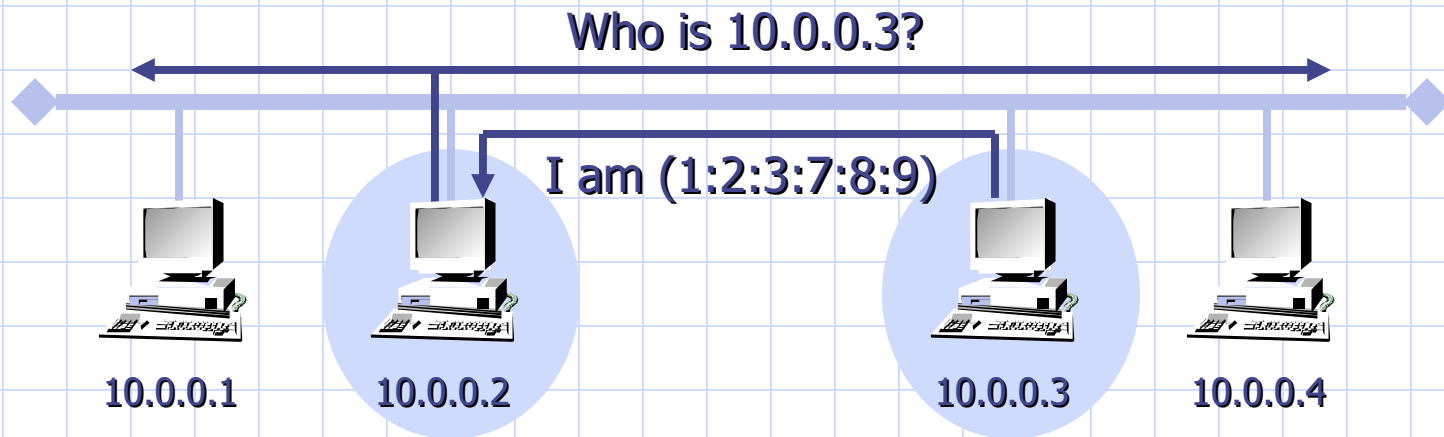


Ethernet Addressing

- ◆ Assigned by manufacturer (hardware)
- ◆ Must be absolutely unique
- ◆ Address format
 - 6 octets in hex (`#: #: #: #: #: #:`)
 - First 3 octets: Manufacturer Identifier
 - Last 3 octets: Card serial number
- ◆ Used for local network communication

Address Resolution Protocol

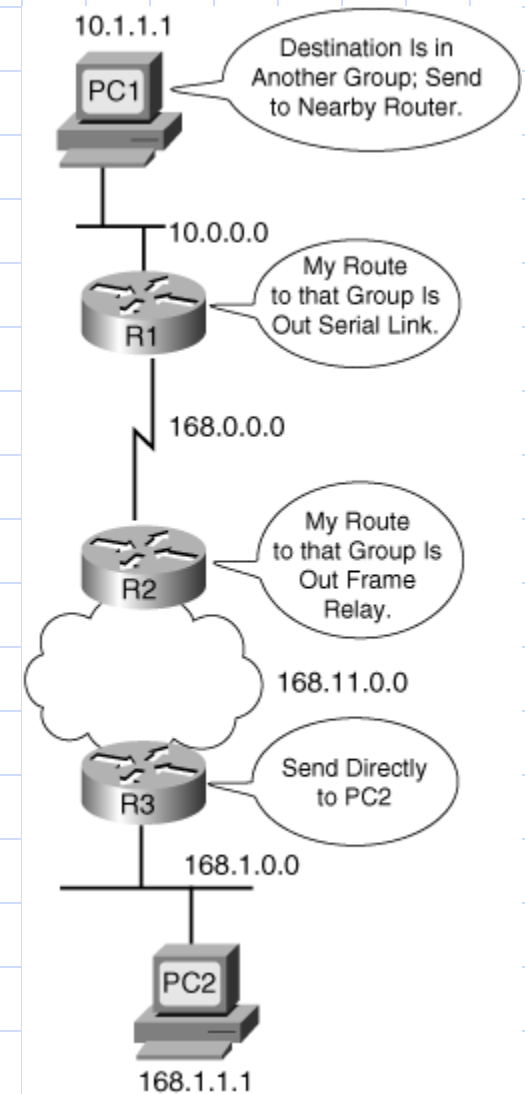
- ◆ Translates IP addresses to Ethernet (MAC) addresses



- ◆ `arp -a`: View the cache

Routing

- ◆ The primary function of the network layer is routing.
- ◆ IP is the widely used network layer protocol.
- ◆ IP routing relies on the structure of IP addresses.

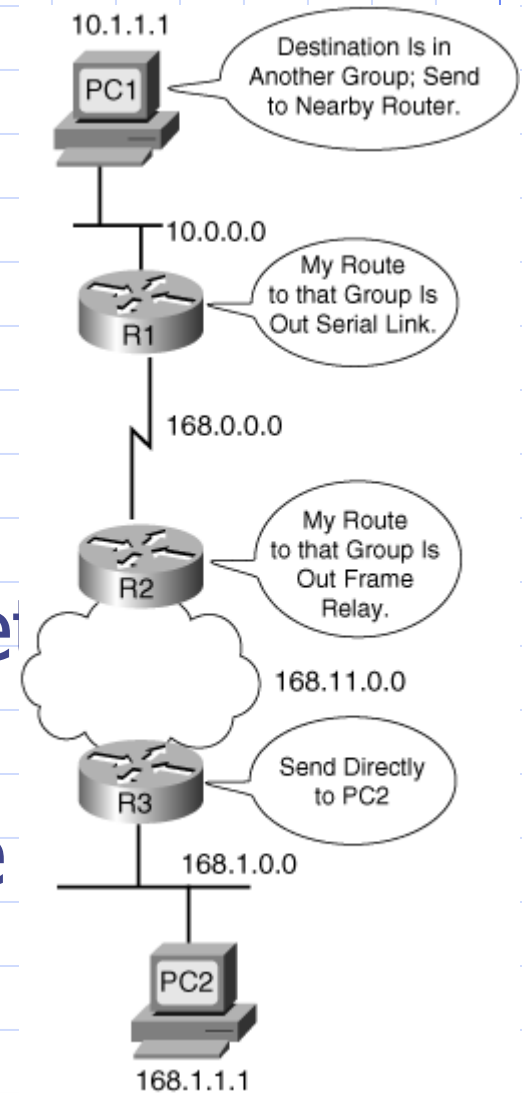


Routing Logic

PC1: If host on same Ethernet, send; otherwise, send to gateway router.

R1, R2: Lookup destination in route table and forward packet to specified router.

R3: Lookup destination in route table and send to local Ethernet.



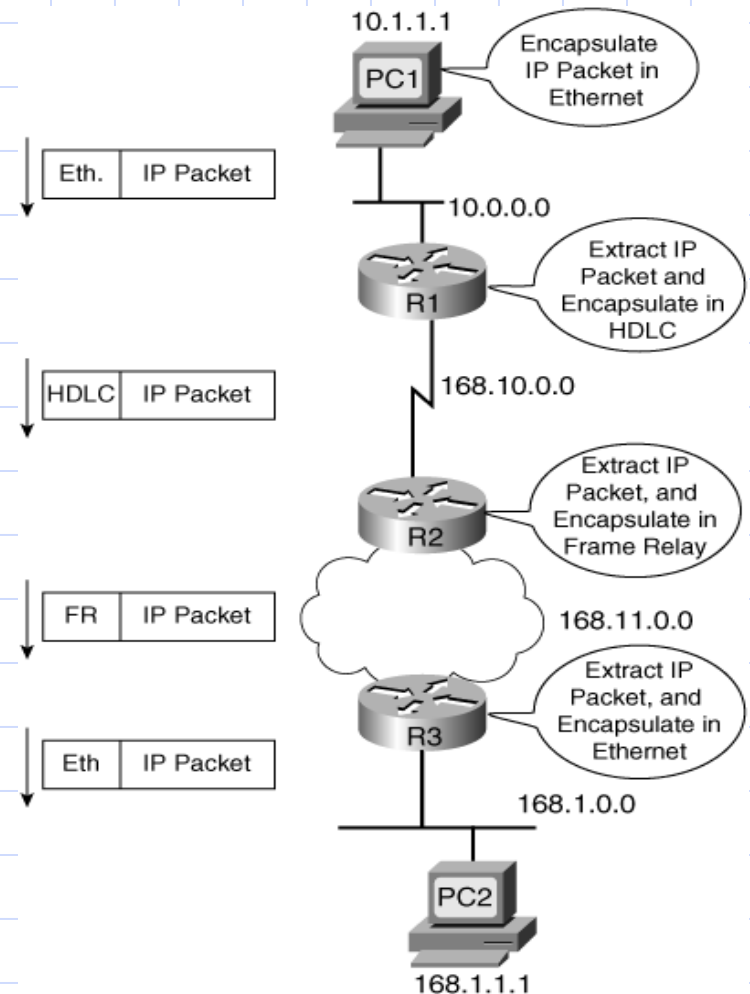
Network Layer Interaction with Data Link Layer

PC1: Network layer encapsulates data in Ethernet frame.

R1: Extract IP packet from Ethernet frame, and encapsulate in HDLC frame to send to R2.

R2: Extract IP packet from HDLC frame and encapsulate in FR frame.

R3: Extract IP packet from FR frame and encapsulate in Ethernet frame.

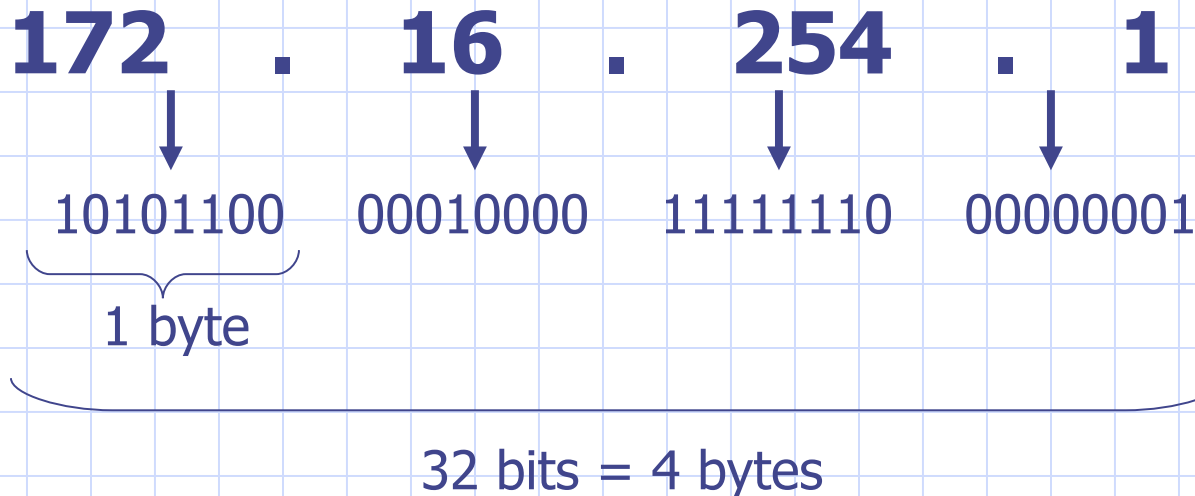


IP Addresses

32-bit integers

One for each network interface.

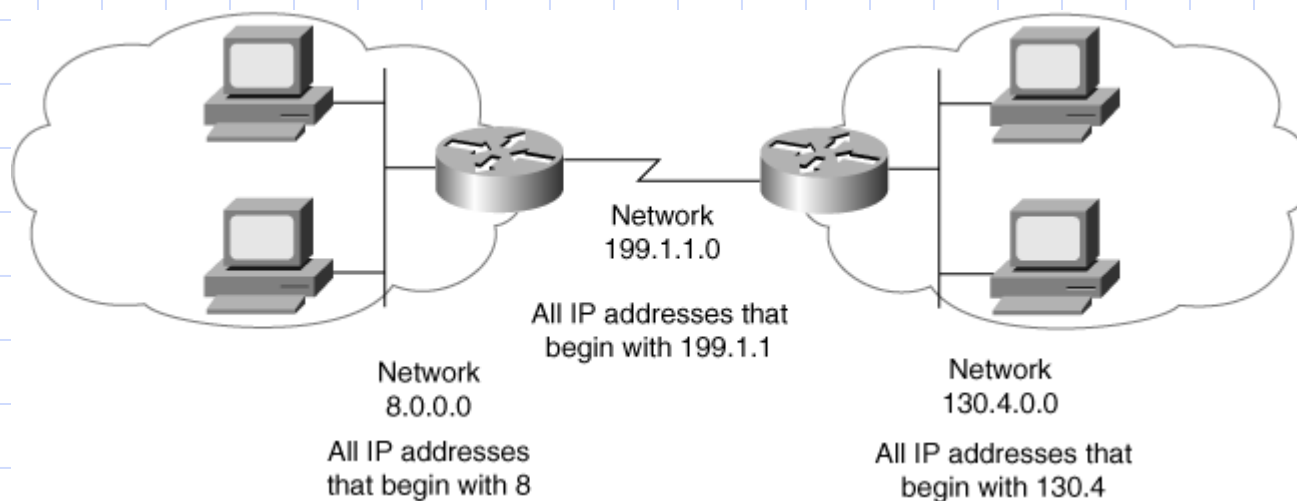
Dotted decimal notation: ii.jj.kk.ll



Grouping IP Addresses

Groups of consecutive IP addrs are called ***networks***.

Routing table would only need 3 entries below.



Network and Host Parts

IP addresses are divided into two parts

- Network ID (like zip code)
- Host ID (like street address)

Network ID	Host ID
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Two special IP addresses

- Network address (e.g. 130.4.0.0)
- Broadcast address (e.g. 130.4.255.255)

Router Logic

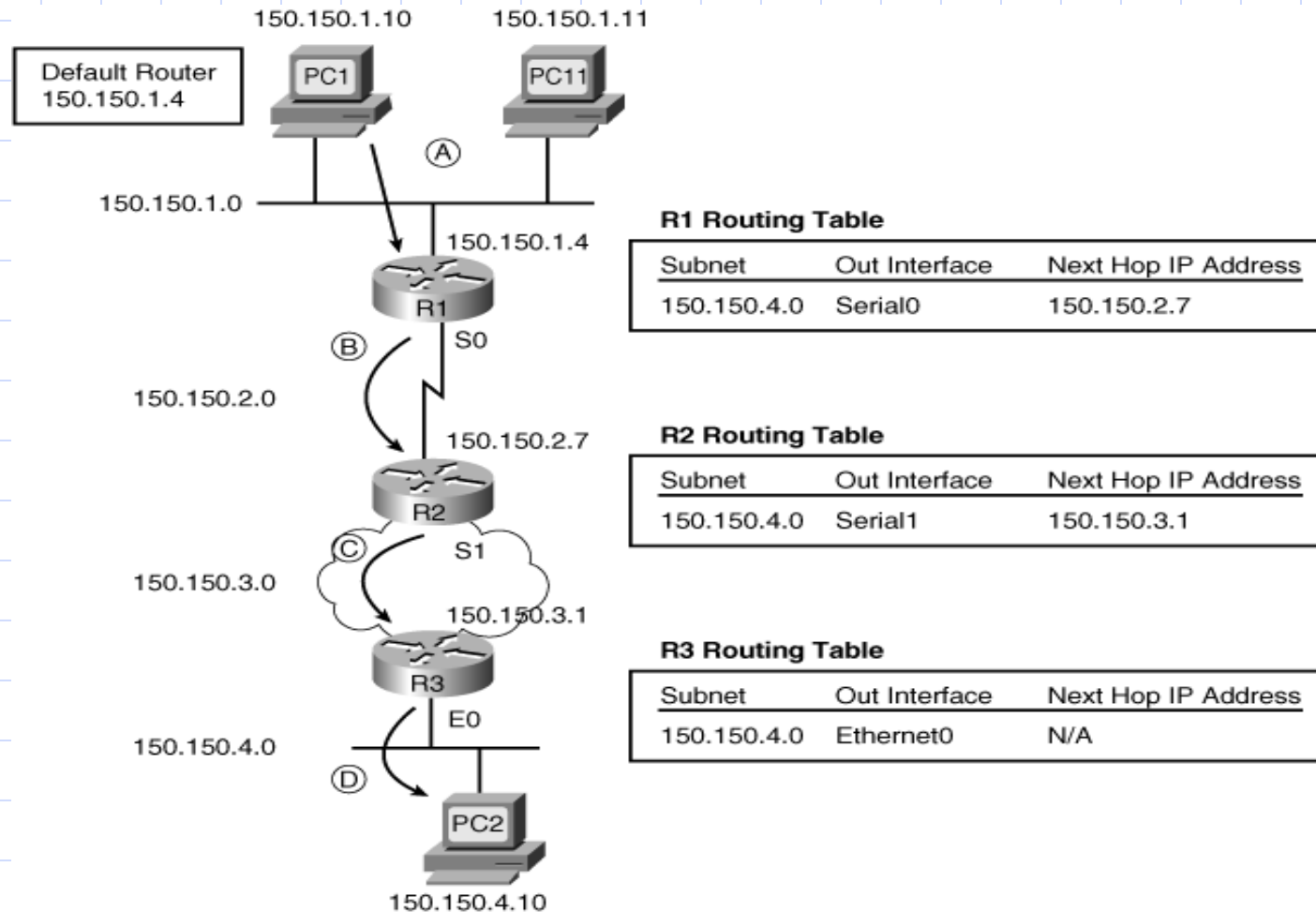
1. Check data-link FCS field for errors. Discard packet if an error detected.
2. Extract IP packet and discard data link header and trailer.
3. Compare packet's destination IP address with routing table, and find route that matches the address. This route identifies the outgoing interface of router.
4. Encapsulate IP packet inside a new data link header and trailer appropriate for outgoing interface, and forward the frame.

Routing Example

PC1 sending a packet to PC2.

1. PC1 to R1.
2. R1 to R2.
3. R2 to R3.
4. R3 to PC2.

Routing Example



Routing Table

Where to send an IP packet to?

Use a table lookup: routing table

Search Process:

1. Search for a matching host address.
2. Search for a matching network address.
3. Search for a default route.

No route to destination: Host or network unreachable error if search fails.

Routing Table

```
st 361m13 ( 10. 1. 0. 90) > net stat -r n
```

Routing Table: IPv4

Destination	Gateway	Flags	Ref	Use	Int
-----	-----	-----	-----	-----	-----
10. 1. 0. 0	10. 1. 0. 90	U	1	4977	hme0
224. 0. 0. 0	10. 1. 0. 90	U	1	0	hme0
defaul t	10. 1. 0. 1	UG	1	66480	
127. 0. 0. 1	127. 0. 0. 1	UH	6	798905	lo0

Routing Table

Destination: final destination host/network

Gateway: next host in route to destination

Flags

U: Route is up

G: Route is to a gateway (router)

H: Route destination is a host (not a network)

D: Route created by a redirect

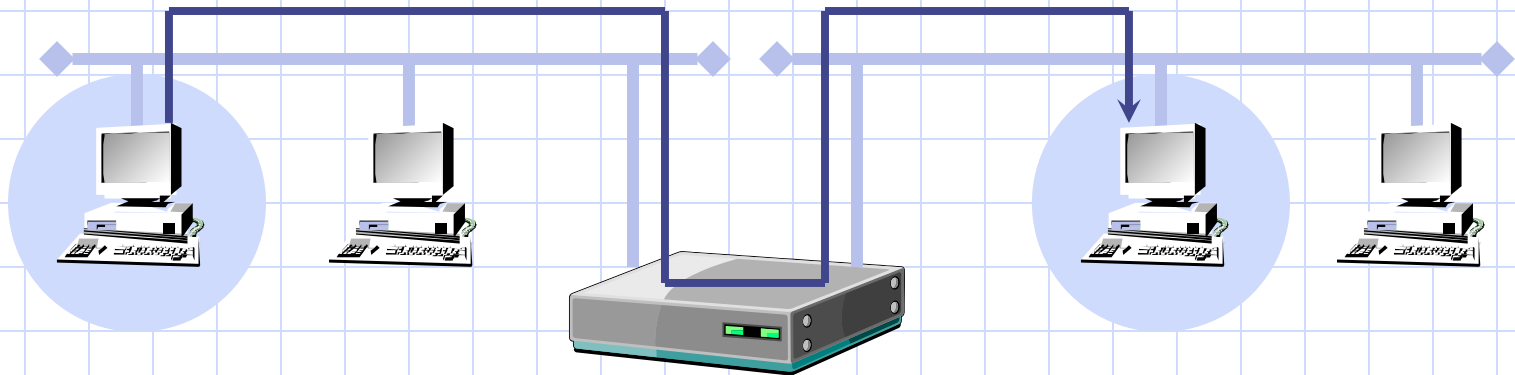
M: Route modified by a redirect

IP Routing Protocol Goals

1. Dynamically learn and fill routing table with a route to all subnets in network.
2. If more than one route available, place the best route in the routing table.
3. Replace lost routes with the best available route as soon as possible.
4. Prevent routing loops.

Default Gateways

- ◆ Connects Networks together
- ◆ If destination not on local network, packets sent through gateway



- ◆ **route:** Display/configure routing

RedHat Network Files

◆ /etc/sysconfig/network

HOSTNAME

◆ /etc/sysconfig/network-scripts/
ifcfg-[interface]

BOOTPROTO

ONBOOT

IPADDR

GATEWAY

USERCTL

NETMASK

NETWORK

BROADCAST

◆ ifup/ifdown [interface]

Name Services

◆ `/etc/hosts`

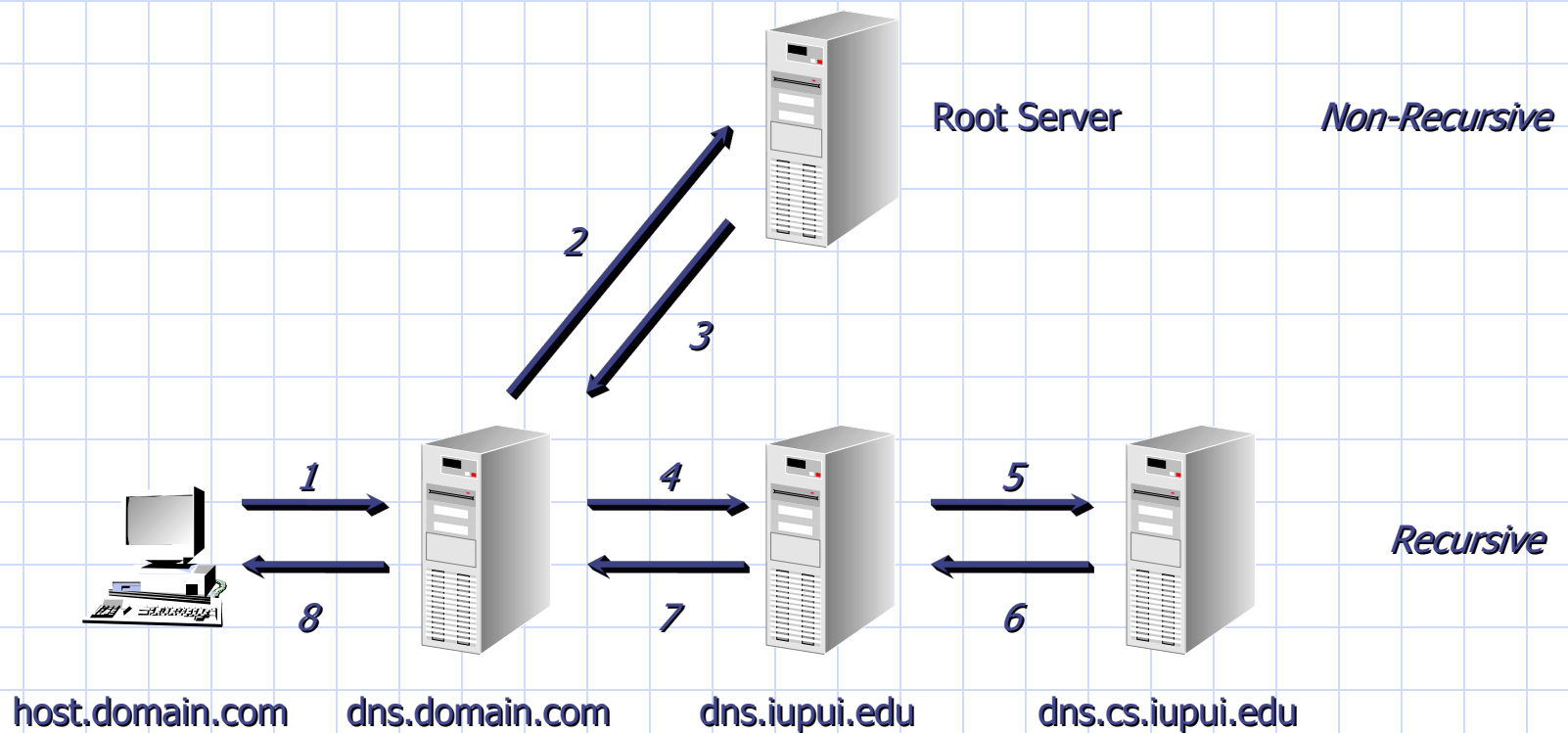
- Local configuration
- Localhost – 127.0.0.1

◆ `/etc/resolv.conf`

- Domain Name Service (DNS) lookup
- `search`: domains to search if not FQDN
- `nameserver` (3): Nameservers to consult

◆ `/etc/nsswitch.conf`

DNS Name Resolution



Network Testing

- ◆ Localhost reachability
- ◆ Hostname reachability
- ◆ Local network reachability
- ◆ Internet network reachability
- ◆ DNS resolution

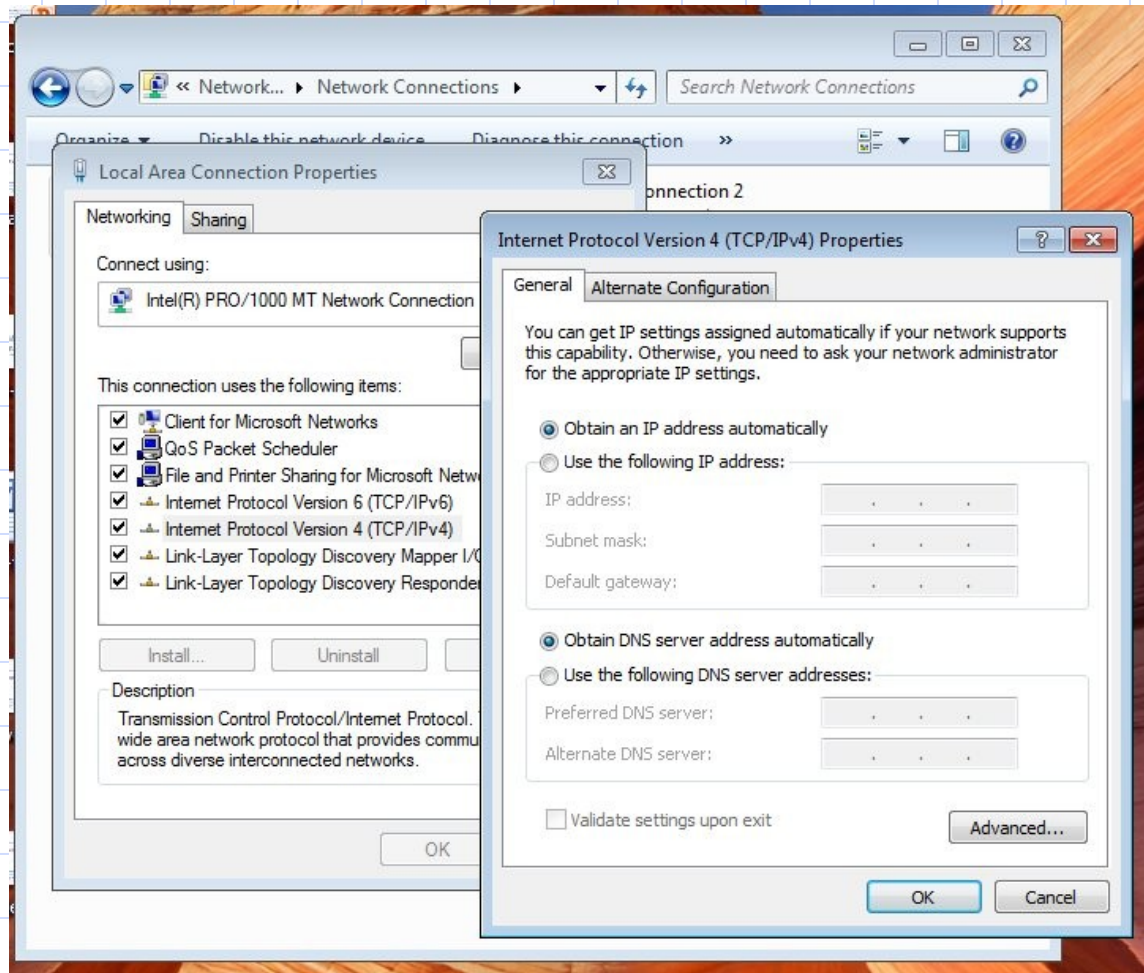
Network tools

- ◆ `ping` – Reachability test
- ◆ `traceroute` – Routing performance
- ◆ `netstat` – Network performance stats
- ◆ `tcpdump` – Packet sniffing
- ◆ `nslookup/dig` – DNS Queries
- ◆ Configuration tools (already discussed)

Dynamic Host Config Protocol

- ◆ Client broadcasts a request for an IP address and network information
- ◆ Server *leases* address to client
- ◆ Lease must be renewed periodically
- ◆ Easy to make global network changes
- ◆ Linux: `BOOTPROTO=dhcp`

Windows Networking



Windows CLI

- ◆ `ipconfig` – Display Interface Settings
- ◆ `ping` – Destination reachability
- ◆ `tracert` – Router hops to destination
- ◆ `netstat` – Performance statistics
- ◆ `nslookup` – DNS lookups
- ◆ `route` – Set/Display gateway
- ◆ `netsh` – Change Interface Settings

netsh Examples

◆ Display Interfaces

```
netsh interface show interface
```

◆ Configure Interface

```
netsh interface ip set address \  
local static [ip-addr] [netmask] \  
[default-gw] 1
```

◆ DNS Server Setting

```
netsh interface ip set dns local \  
static [ip-addr]
```

Wireless Networks

- ◆ Extend the network
- ◆ Included in many devices now
 - Laptops
 - Smart Phones
 - DSL/Cable Modems
- ◆ Bandwidth (YMMV!)
 - 802.11b – 11 Mbps
 - 802.11g – 54 Mbps
 - 802.11n - 150/300/450/600 Mbps
- ◆ Set Service Identifier (SSID)
 - Shared “key” between clients and Access Point (AP)
 - Automatically detected vs. assigned

Wireless Security Issues

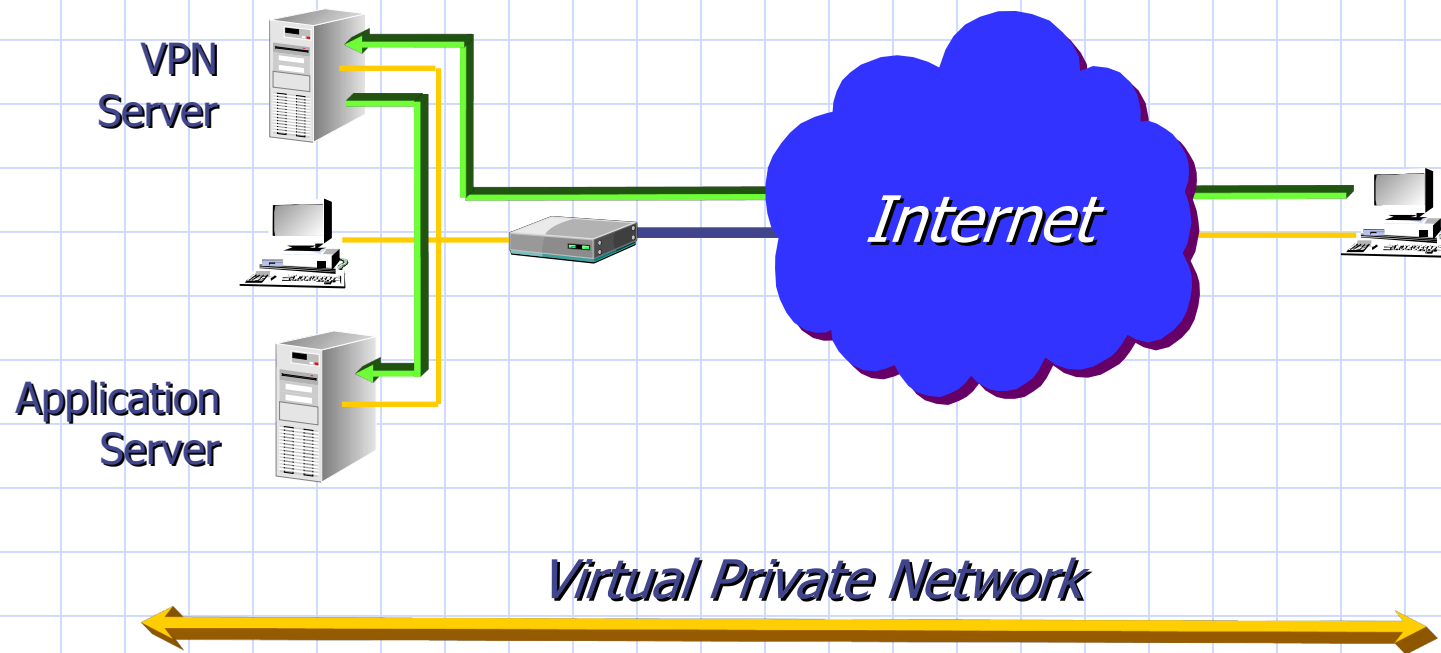
- ◆ Sniffing / War Driving
- ◆ Bandwidth stealing
- ◆ Access to private resources
- ◆ Security Measures
 - Non-broadcasting SSIDs
 - MAC Access Control Lists (ACLs)
 - WEP????
 - WPA/WPA2

RedHat ifcfg- Additions

- ◆ `TYPE=Wireless`
- ◆ `ESSID=[ssid name]`
- ◆ `CHANNEL=[1-11]`
- ◆ `MODE=[Auto | Managed | Ad-hoc]`

Can set manually with `/sbin/iwconfig`

Virtual Private Networks



Point to Point Tunneling Protocol

◆ Based on

- Point to Point Protocol (PPP)
- Generic Routing Encapsulation (GRE)



■ Weaknesses

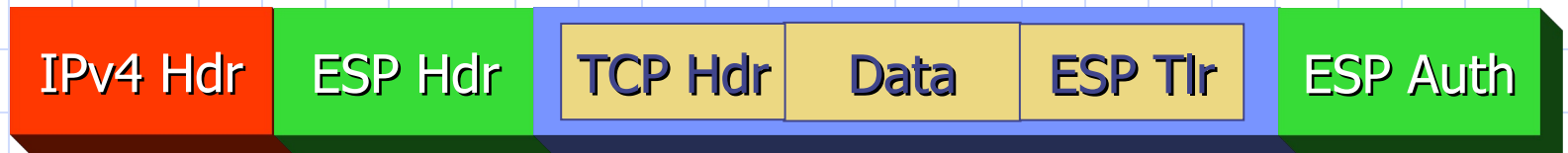
- Poor Encryption
- Session handshaking done in clear

IPSec

- ◆ Part of IPv6 Spec
- ◆ Authentication Header (AH)



- Encapsulating Security Payload (ESP)



- Modes: Transport and Tunnel